MITIGATING REASONING HALLUCINATIONS IN LLMS VIA NEURO-SYMBOLIC INTEGRATION FOR ILL-DEFINED MATHEMATICAL PROBLEMS

Jave Bacsain
javbacsain@my.cspc.edu.ph

Ivy Pauline Muit
ivmuit@my.cspc.edu.ph

The Problem

Large Language Models (LLMs) frequently produce confident yet incorrect answers to math problems that are missing key information or contain contradictory conditions — a failure known as hallucination. Standard LLMs will still generate a numeric answer even when no valid solution exists, which is dangerous in high-stakes settings like medical dosing, financial systems, or automated grading.

Objectives

Design a system that either:

- Returns a numeric answer if the problem is solvable, or
- Returns a REJECT label if the problem is ill-defined (missing info or contradictory)

Key Findings & Results

- The pipeline reduced hallucination on ill-defined problems from ~20% to 2% — a ~10× reduction
- It is the only system to correctly detect contradictions (baselines score F1=0.000 on contra class)
- The main trade-off: the 7B translator under-constrains SMT-LIB on ~94% of solvable inputs, causing over-abstention and lower macro F1
- In the binary reject-vs-solve view, the pipeline's F1 (0.799) is only 8% behind the best baseline (0.880)

Conclusion

Separating language understanding (LLM) from logical evaluation (Z3 solver) eliminates a class of hallucinations that prompting alone cannot avoid. While aggregate accuracy trails the baselines, the pipeline's near-zero hallucination rate on ill-defined problems validates the core hypothesis. Future work includes fine-tuning the translator and extending to typed SMT-LIB for unit-aware reasoning.

CSEC 6 : Natural Language Processing

Methodology

A two-stage neuro-symbolic pipeline:

1

Stage 1: LLM as Translator

- Model: Qwen2.5-7B-Instruct (4-bit quantized)
- Translates math word problems into SMT-LIB v2 formal logic using 10 few-shot demonstrations
- Does NOT solve the problem — only describes its logical structure

2

Step 2: Z3 Constraint Solver as Judge

- **UNSAT** → Contradictory problem → REJECT
- **SAT + unique model** → Solvable → return numeric answer
- **SAT + multiple models** → Missing information → REJECT
- Parse errors / timeouts → REJECT (abstain by default)

3

Dataset: 600-problem balanced test set

- 200 from PMC missing_test split
- 200 from PMC contra_test split
- 200 from GSM8K (solvable)

References

[1] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, et al. Language models are few-shot learners. In Advances in Neural Information Processing Systems 33 (NeurIPS 2020), 2020.

[2] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. ACM Transactions on Information Systems, 43(2):1–55, 2025.

[3] E. Puchalska and Z. Semadeni. Children's reactions to verbal arithmetical problems with missing, surplus or contradictory data. For the Learning of Mathematics, 7(3):9–16, 1987.